

# Write Equations

Flagship

Lesson 7-1-flagship

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

## COLD CASE MISSION

## The Equation Vault

You are Detective Ruiz's new code analyst. A tip just came in: 'A number of stolen gems plus 8 more equals 20 total.' The evidence vault only opens when the clue is translated into a correct equation. Master writing equations and the case cracks wide open.

## Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$x + 5 = 12$  means 'some number plus 5 equals 12'

### Equation

A math sentence with an equal sign showing both sides are the same.

In  $n + 3 = 10$ , the letter  $n$  stands for the unknown number ( $n = 7$ )

### Variable

A letter that stands for an unknown number.

$7 + 3 = 10$  — both sides equal 10

### Equal sign

The = sign, showing both sides are the same.

$2x + 5$  is an expression; it becomes an equation when you write  $2x + 5 = 15$

### Expression

A math phrase with numbers and letters, but no equal sign.

## Key Ideas & Notes

- Detective Ruiz received a tip: 'A number of stolen gems plus 8 more equals 20 total gems.' She needs to write an equation to represent this clue before she can solve the case.
- How would you write this situation as an equation?
- Use the balance scale to see how each word problem translates into an equation. Drag the correct expression to each side of the scale to make it balance.

### Think About It

- What quantity is unknown in this situation?
- What does 'plus 8 more' tell you about the operation?
- What does 'equals 20 total' tell you?

### My Notes

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## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**Detective Ruiz's clue says 'a number of stolen gems plus 8 more equals 20 total.' How would you turn this clue into an equation, and what does each part stand for?**

#### Level 1 support

**Start here:** I let a variable stand for the unknown number of gems.

**The unknown is** \_\_\_\_, **so I write the variable** \_\_\_\_.

*La incógnita es \_\_\_\_, así que escribo la variable \_\_\_\_.*

**My equation is** \_\_\_\_ **because 'plus 8' means** \_\_\_\_ **and 'equals 20' means** \_\_\_\_.

*Mi ecuación es \_\_\_\_ porque 'más 8' significa \_\_\_\_ y 'es igual a 20' significa \_\_\_\_.*

**Word bank:** equation variable equal sign expression

**Listen for:** Students choose a variable (such as  $n$ ) for the unknown gems, build  $n + 8 = 20$ , and connect 'plus 8 more' to  $+ 8$  and 'equals 20 total' to  $= 20$ .

#### Level 2

**Why does the equation  $n + 8 = 20$  represent the clue better than the expression  $n + 8$  alone? Justify using the role of the equal sign.**

- The equation is better than the expression because \_\_\_\_.
- The equal sign tells me \_\_\_\_ that an expression cannot show.

EXPLORE

**On the balance scale you put  $n + 8$  on one side and 20 on the other. How did you decide what belongs on each side of the equal sign?**

Level 1 support

**Start here:** Both sides of the equation must represent the same total.

**I put \_\_\_\_ on the left side because \_\_\_\_.**

*Puse \_\_\_\_ en el lado izquierdo porque \_\_\_\_.*

**The equal sign means both sides are \_\_\_\_.**

*El signo igual significa que ambos lados son \_\_\_\_.*

**Word bank:** equation variable equal sign expression

**Listen for:** Students place the expression  $n + 8$  on one side and the total 20 on the other, explaining the equal sign means both sides have the same value.

Level 2

**Compare 'twice a number equals 20' with 'a number plus 8 equals 20.' How would the equations differ, and why?**

- 'Twice a number' would be \_\_\_\_ instead of \_\_\_\_ because \_\_\_\_.
- The operation changes the equation because \_\_\_\_.

CONNECT

**When would writing an equation help you solve a real problem instead of just guessing?**

Level 1 support

**Start here:** An equation organizes what I know and what I want to find.

**Writing an equation helps me because \_\_\_\_.**

*Escribir una ecuación me ayuda porque \_\_\_\_.*

**I would write an equation to find \_\_\_\_.**

*Escribiría una ecuación para hallar \_\_\_\_.*

**Word bank:** **equation** **variable** **expression** **equal sign**

**Listen for:** *Students explain that an equation captures the relationship clearly, lets you use a variable for the unknown, and can be solved instead of guessing.*

Level 2

**Critique: 'Any sentence with numbers is an equation.' Use the words expression and equal sign to explain why this is wrong.**

- This is wrong because an expression \_\_\_\_.
- A sentence is only an equation when it has \_\_\_\_.

## Guided Examples

### Example 1

**Which equation represents: 'A number plus 12 equals 30'?**

**Solution:** 'Plus' means addition, so the equation is  $n + 12 = 30$ .

**Answer:** A.  $n + 12 = 30$

### Example 2

**Which equation represents: 'Seven less than a number is 18'?**

**Solution:** 'Seven less than a number' means start with the number and subtract 7:  $n - 7 = 18$ .

**Answer:** A.  $n - 7 = 18$

### Example 3

**Which equation represents: 'Three times a number equals 21'?**

**Solution:** 'Three times' means multiply by 3, so the equation is  $3n = 21$ .

**Answer:** A.  $3n = 21$

## Write About the Math

### The Writing Revolution

I can explain my equation using the words equation, variable, equal sign, and expression.

#### 1. Kernel Sentence subject + verb

**Model:** Equation is a math sentence with an equal sign showing both sides are the same.  
*Ecuación es una oración matemática con un signo igual que muestra que ambos lados son iguales.*

**Write a kernel sentence about equation. Use a subject and a verb.**

*Escribe una oración base sobre ecuación. Usa un sujeto y un verbo.*

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#### 2. Sentence Expansion because · but · so

**Kernel:** Equation matters in math  
*Ecuación importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Equation matters in math because \_\_\_\_.**  
*Ecuación importa en matemáticas porque \_\_\_\_.*

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**but**  
*pero*

**Equation matters in math, but \_\_\_\_.**  
*Ecuación importa en matemáticas, pero \_\_\_\_.*

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**so**  
*entonces*

**Equation matters in math, so \_\_\_\_.**  
*Ecuación importa en matemáticas, entonces \_\_\_\_.*

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### 3. Sentence Types 4 ways to write a math idea

**Statement**  
*Afirmación*

Tell one true fact about equation.  
*Di un hecho verdadero sobre equation.*

**Equation** \_\_\_\_.

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**Question**  
*Pregunta*

Ask a question about equation.  
*Haz una pregunta sobre equation.*

**How does** \_\_\_\_ ?

*¿Cómo* \_\_\_\_ ?

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**Exclamation**  
*Exclamación*

Show excitement about equation.  
*Muestra entusiasmo sobre equation.*

**Wow,** \_\_\_\_ !

*¡Guau,* \_\_\_\_ !

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**Command**  
*Mandato*

Tell a partner what to do with equation.  
*Dile a un compañero qué hacer con equation.*

**First,** \_\_\_\_ .

*Primero,* \_\_\_\_ .

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### 4. Explain Your Reasoning use a sentence starter

**The unknown is** \_\_\_\_, **so I write the variable** \_\_\_\_.

*La incógnita es* \_\_\_\_, *así que escribo la variable* \_\_\_\_.

**My equation is** \_\_\_\_ **because 'plus 8' means** \_\_\_\_ **and 'equals 20' means** \_\_\_\_.

*Mi ecuación es* \_\_\_\_ *porque 'más 8' significa* \_\_\_\_ *y 'es igual a 20' significa* \_\_\_\_.

**I put** \_\_\_\_ **on the left side because** \_\_\_\_.

*Puse* \_\_\_\_ *en el lado izquierdo porque* \_\_\_\_.

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## Try It

Solve on your own. Check the answer key when you are done.

**1. Which equation represents: 'A number divided by 4 equals 8'?**

A.  $n / 4 = 8$

B.  $4n = 8$

C.  $n - 4 = 8$

D.  $n + 4 = 8$

Show your work:

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**2. A detective found some fingerprints on Monday and 14 more on Tuesday, for a total of 30. Which equation represents this?**

A.  $f + 14 = 30$

B.  $f - 14 = 30$

C.  $14f = 30$

D.  $f / 14 = 30$

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**Write a real-world situation that can be modeled by the equation  $n / 5 = 9$ . Explain what  $n$  represents and verify your equation makes sense.**

*Sentence starter: Situation: \_\_\_\_\_. In this problem,  $n$  represents \_\_\_\_\_. Check:  $n = \underline{\hspace{1cm}}$  because  $\underline{\hspace{1cm}} \div 5 = 9$ .*

Show your work:

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## Reflect — Exit Ticket

**Which equation represents: 'A number divided by 6 equals 9'?**

- A.  $n / 6 = 9$
- B.  $6n = 9$
- C.  $n - 6 = 9$
- D.  $n + 6 = 9$

Your answer:

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## Answer Key & Teacher Guide

1. **Try It 1:**  $A. n / 4 = 8$  — *'Divided by 4' means division:  $n / 4 = 8$ .*
2. **Try It 2:**  $A. f + 14 = 30$  — *The detective's Monday fingerprints ( $f$ ) plus 14 more equals 30:  $f + 14 = 30$ .*
3. **Exit Ticket:**  $A. n / 6 = 9$  — *'Divided by 6' means division:  $n / 6 = 9$ . Check:  $54 / 6 = 9$  ✓*

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Equation is a math sentence with an equal sign showing both sides are the same.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).